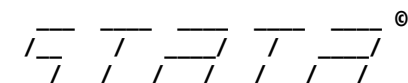


User: PLFS Karnataka Results PDF Output
Project: Linear Regression Models



Statistics and Data Science

17.0
MP-Parallel Edition

Copyright 1985-2021 StataCorp LLC
StataCorp
4905 Lakeway Drive
College Station, Texas 77845 USA
800-STATA-PC <https://www.stata.com>
979-696-4600 stata@stata.com

Stata license: Unlimited-user 64-core network perpetual
Serial number: 18461036
Licensed to: TEAM BTCR
TEAM BTCR

Notes:

1. Unicode is supported; see [help unicode advice](#).
2. More than 2 billion observations are allowed; see [help obs advice](#).
3. Maximum number of variables is set to 5,000; see [help set maxvar](#).

```
1 . do "C:\Users\Tharun\Downloads\PLFS Data July 2023 to June 2024\PLFS - Regressions.do"

2 . use "C:\Users\Tharun\Downloads\PLFS Data July 2023 to June 2024\DDI-IND-CSO-PLFS-2023-24 - Stata Files\Fully Me
> "

3 . * --- Step 1.1: Calculate Weekly Wage - Activity 1 (Sum across 7 days) ---
4 . gen double weekly_wage_act1 = 0

5 . * Day 7 (Suffix 1)
6 . replace weekly_wage_act1 = weekly_wage_act1 + b6q9_3pt1_perrv if inlist(b6q4_3pt1_perrv, "31", "71", "72")
   (0 real changes made)

7 . * Day 6 (Suffix 2) - Correcting for status variable anomaly (b6q4_3pt2)
8 . replace weekly_wage_act1 = weekly_wage_act1 + b6q9_3pt2_perrv if inlist(b6q4_3pt2_perrv, "31", "71", "72")
   (0 real changes made)

9 . * Day 5 (Suffix 3)
10 . replace weekly_wage_act1 = weekly_wage_act1 + b6q9_3pt3_perrv if inlist(b6q4_3pt3_perrv, "31", "71", "72")
    (0 real changes made)

11 . * Day 4 (Suffix 4)
12 . replace weekly_wage_act1 = weekly_wage_act1 + b6q9_3pt4_perv1 if inlist(b6q4_3pt4_perrv, "31", "71", "72")
    (85,135 real changes made, 85,135 to missing)

13 . * Day 3 (Suffix 5)
14 . replace weekly_wage_act1 = weekly_wage_act1 + b6q9_3pt5_perv1 if inlist(b6q4_3pt5_perrv, "31", "71", "72")
    (250 real changes made, 250 to missing)

15 . * Day 2 (Suffix 6)
```

```
16 . replace weekly_wage_act1 = weekly_wage_act1 + b6q9_3pt6_perv1 if inlist(b6q4_3pt6_perrv, "31", "71", "72")
    (146 real changes made, 146 to missing)

17 . * Day 1 (Suffix 7)
18 . replace weekly_wage_act1 = weekly_wage_act1 + b6q9_3pt7_perv1 if inlist(b6q4_3pt7_perrv, "31", "71", "72")
    (108 real changes made, 108 to missing)

19 . * --- Step 1.2: Calculate Weekly Wage - Activity 2 (Sum across 7 days) ---
20 . gen double weekly_wage_act2 = 0

21 . * Day 7 (Suffix 1)
22 . replace weekly_wage_act2 = weekly_wage_act2 + b6q9_act2_3pt1_perv1 if inlist(b6q4_act2_3pt1_perrv, "31", "71",
    (200 real changes made, 200 to missing)

23 . * Day 6 (Suffix 2)
24 . replace weekly_wage_act2 = weekly_wage_act2 + b6q9_act2_3pt2_perv1 if inlist(b6q4_act2_3pt2_perrv, "31", "71",
    (20 real changes made, 20 to missing)

25 . * Day 5 (Suffix 3)
26 . replace weekly_wage_act2 = weekly_wage_act2 + b6q9_act2_3pt3_perv1 if inlist(b6q4_act2_3pt3_perrv, "31", "71",
    (17 real changes made, 17 to missing)

27 . * Day 4 (Suffix 4)
28 . replace weekly_wage_act2 = weekly_wage_act2 + b6q9_act2_3pt4_perv1 if inlist(b6q4_act2_3pt4_perrv, "31", "71",
    (1 real change made, 1 to missing)

29 . * Day 3 (Suffix 5)
30 . replace weekly_wage_act2 = weekly_wage_act2 + b6q9_act2_3pt5_perv1 if inlist(b6q4_act2_3pt5_perrv, 31, 71, 72)
    (3 real changes made, 3 to missing)

31 . * Day 2 (Suffix 6)
32 . replace weekly_wage_act2 = weekly_wage_act2 + b6q9_act2_3pt6_perv1 if inlist(b6q4_act2_3pt6_perrv, "31", "71",
    (5 real changes made, 5 to missing)

33 . * Day 1 (Suffix 7)
34 . replace weekly_wage_act2 = weekly_wage_act2 + b6q9_act2_3pt7_perrv if inlist(b6q4_act2_3pt7_perrv, "31", "71",
    (0 real changes made)

35 . * --- Step 1.3: Total Monthly Earnings and Log Transformation ---
36 . gen double total_monthly_earnings = (weekly_wage_act1 + weekly_wage_act2) * 4 + b6q9_perv1
    (504,440 missing values generated)

37 . * --- Age and Age Squared ---
38 . rename b4q6_perv1 age

39 . label var age "Age"

40 . gen double age_sq = age^2
    (504,440 missing values generated)

41 . label var age_sq "Age Squared"
```

```

42 . * --- Education ---
43 . rename b4q10_perv1 education

44 . label var education "No. of years in Formal Education"

45 . gen double edu_sq = education^2
    (504,440 missing values generated)

46 . label var edu_sq "Education Squared"

47 . * --- Urban Dummy (Urban=1, Rural=0) ---
48 . * Sector: 1=Rural, 2=Urban
49 . gen urban = (Sector == "2") if !missing(Sector)

50 . label var urban "Urban (1=Urban, 0=Rural)"

51 . * --- Gender Dummy (Female=1, Male=0) ---
52 . * Gender (b4q5_perv1): 1=Male, 2=Female
53 . gen female = (b4q5_perv1 == "2") if !missing(b4q5_perv1)
    (504,440 missing values generated)

54 . label var female "Female (1=Female, 0=Male)"

55 . display _N
    922599

56 . count if state_hhv1 == "29"
    15,054

57 . summarize total_monthly_earnings, detail

```

total_monthly_earnings				
	Percentiles	Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	418,159
25%	0	0	Sum of wgt.	418,159
50%	0		Mean	2403.436
		Largest	Std. dev.	9482.504
75%	0	300000		
90%	6000	300000	Variance	8.99e+07
95%	15500	300000	Skewness	6.908731
99%	48500	300000	Kurtosis	83.79505

```

58 . drop if total_monthly_earnings <= 0 | missing(total_monthly_earnings)
    (876,305 observations deleted)

59 . gen double ln_earnings = log(total_monthly_earnings)

```

```
60 . label var ln_earnings "Ln(Total Monthly Earnings)"
```

```
61 . display _N
46294
```

```
62 . count if state_hhv1 == "29"
1,919
```

```
63 . summarize ln_earnings, detail
```

Ln(Total Monthly Earnings)				
	Percentiles	Smallest		
1%	7.600902	5.991465		
5%	8.429454	6.214608		
10%	8.699515	6.214608	Obs	46,294
25%	9.159047	6.214608	Sum of wgt.	46,294
50%	9.615805		Mean	9.668863
		Largest	Std. dev.	.7978623
75%	10.23996	12.61154		
90%	10.7364	12.61154	Variance	.6365843
95%	11.0021	12.61154	Skewness	-.0291992
99%	11.40756	12.61154	Kurtosis	3.258161

```
64 . regress ln_earnings age_sq edu_sq urban female age education
```

Source	SS	df	MS	Number of obs	=	46,294
Model	12429.6531	6	2071.60885	F(6, 46287)	=	5627.35
Residual	17039.7442	46,287	.368132397	Prob > F	=	0.0000
				R-squared	=	0.4218
				Adj R-squared	=	0.4217
Total	29469.3973	46,293	.636584307	Root MSE	=	.60674

ln_earnings	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age_sq	-.0007193	.0000191	-37.66	0.000	-.0007567	-.0006818
edu_sq	.003609	.0001084	33.31	0.000	.0033966	.0038214
urban	.1891324	.0060091	31.47	0.000	.1773545	.2009103
female	-.4731272	.0065746	-71.96	0.000	-.4860136	-.4602408
age	.0746446	.0015268	48.89	0.000	.071652	.0776371
education	.0169417	.0022029	7.69	0.000	.012624	.0212593
_cons	7.263716	.0311266	233.36	0.000	7.202707	7.324724

```
65 . keep if state_hhv1 == "29"
(44,375 observations deleted)
```

```
66 . display _N
1919
```

67 . * Model 1: Earnings as a Function of Age and Education

68 . regress total_monthly_earnings age education, robust

Linear regression	Number of obs	=	1,919
	F(2, 1916)	=	317.27
	Prob > F	=	0.0000
	R-squared	=	0.3232
	Root MSE	=	18029

total_monthly_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	483.1076	50.61387	9.54	0.000	383.8435	582.3716
education	2842.621	118.0099	24.09	0.000	2611.179	3074.062
_cons	-25701.75	2667.494	-9.64	0.000	-30933.25	-20470.26

69 . * Model 2: Earnings with Age, Education, and Gender

70 . regress total_monthly_earnings age education female, robust

Linear regression	Number of obs	=	1,919
	F(3, 1915)	=	213.67
	Prob > F	=	0.0000
	R-squared	=	0.3327
	Root MSE	=	17907

total_monthly_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	488.8674	50.85414	9.61	0.000	389.1321	588.6027
education	2780.532	116.6546	23.84	0.000	2551.749	3009.315
female	-4679.045	820.6389	-5.70	0.000	-6288.485	-3069.605
_cons	-23756.63	2587.212	-9.18	0.000	-28830.68	-18682.59

71 . * Model 3: Earnings with Age, Education, and Sector

72 . regress total_monthly_earnings age education urban, robust

Linear regression	Number of obs	=	1,919
	F(3, 1915)	=	225.28
	Prob > F	=	0.0000
	R-squared	=	0.3434
	Root MSE	=	17763

total_monthly_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	450.1971	50.20356	8.97	0.000	351.7377	548.6565
education	2758.615	114.8183	24.03	0.000	2533.433	2983.797
urban	7781.089	663.3685	11.73	0.000	6480.089	9082.09
_cons	-29710.15	2725.997	-10.90	0.000	-35056.38	-24363.92

73 . * Model 4: Log Wages with Age and Education

74 . regress ln_earnings age education, robust

Linear regression	Number of obs	=	1,919
	F(2, 1916)	=	636.86
	Prob > F	=	0.0000
	R-squared	=	0.4405
	Root MSE	=	.55492

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0142759	.0012084	11.81	0.000	.011906	.0166458
education	.1141285	.0032416	35.21	0.000	.1077709	.120486
_cons	7.99646	.0624693	128.01	0.000	7.873945	8.118975

75 . * Model 5: Log Wages with Age, Education, and Gender

76 . regress ln_earnings age education female, robust

Linear regression	Number of obs	=	1,919
	F(3, 1915)	=	494.50
	Prob > F	=	0.0000
	R-squared	=	0.4778
	Root MSE	=	.53625

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0146627	.0012013	12.21	0.000	.0123067	.0170187
education	.1099593	.0031881	34.49	0.000	.1037067	.1162118
female	-.3141933	.0291731	-10.77	0.000	-.3714077	-.256979
_cons	8.127073	.0619021	131.29	0.000	8.005671	8.248476

77 . * Model 6: Log Wages with Age, Education, and Sector

78 . regress ln_earnings age education urban, robust

Linear regression	Number of obs	=	1,919
	F(3, 1915)	=	466.13
	Prob > F	=	0.0000
	R-squared	=	0.4724
	Root MSE	=	.539

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0128748	.0011956	10.77	0.000	.01053	.0152197
education	.1105521	.0031882	34.67	0.000	.1042993	.1168049
urban	.3312622	.0300257	11.03	0.000	.2723757	.3901487
_cons	7.825812	.0635095	123.22	0.000	7.701257	7.950367

79 . * Model 7: Log Wages with Age and Age² (Life-Cycle Hypothesis)

80 . regress ln_earnings age age_sq, robust

Linear regression	Number of obs	=	1,919
	F(2, 1916)	=	28.31
	Prob > F	=	0.0000
	R-squared	=	0.0255
	Root MSE	=	.73235

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0704799	.0108275	6.51	0.000	.0492451	.0917148
age_sq	-.0008447	.0001428	-5.91	0.000	-.0011247	-.0005646
_cons	8.509933	.1904709	44.68	0.000	8.13638	8.883485

81 . * Model 8: Log Wages with Education and Education² (Gary Becker's Hypothesis)

82 . regress ln_earnings education edu_sq, robust

Linear regression	Number of obs	=	1,919
	F(2, 1916)	=	621.27
	Prob > F	=	0.0000
	R-squared	=	0.4311
	Root MSE	=	.55956

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
education	.0049575	.0114792	0.43	0.666	-.0175556	.0274707
edu_sq	.0051299	.0005708	8.99	0.000	.0040105	.0062494
_cons	8.98563	.055885	160.79	0.000	8.876028	9.095232

83 . * Model 9: Log Wages with Gender, Sector, and Interaction

84 . regress ln_earnings i.female##i.urban, robust

Linear regression	Number of obs	=	1,919
	F(3, 1915)	=	127.07
	Prob > F	=	0.0000
	R-squared	=	0.1435
	Root MSE	=	.68676

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.female	-.5550642	.0650331	-8.54	0.000	-.6826073	-.4275211
1.urban	.4439106	.03806	11.66	0.000	.3692672	.518554
female#urban 1 1	.1380712	.0781075	1.77	0.077	-.0151134	.2912559
_cons	9.636812	.0321742	299.52	0.000	9.573712	9.699912

85 . margins female#urban

Adjusted predictions

Number of obs = 1,919

Model VCE: Robust

Expression: Linear prediction, predict()

	Delta-method					
	Margin	std. err.	t	P> t	[95% conf. interval]	
female#urban						
0 0	9.636812	.0321742	299.52	0.000	9.573712	9.699912
0 1	10.08072	.0203318	495.81	0.000	10.04085	10.1206
1 0	9.081747	.0565165	160.69	0.000	8.970907	9.192588
1 1	9.663729	.038185	253.08	0.000	9.588841	9.738618

86 . * Model 10: Log Wages with Age, Education, Gender, Sector, and Interaction

87 . regress ln_earnings age education i.female##i.urban, robust

Linear regression

Number of obs = 1,919
 F(5, 1913) = 339.48
 Prob > F = 0.0000
 R-squared = 0.5130
 Root MSE = .51813

ln_earnings	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0133221	.001183	11.26	0.000	.0110019	.0156423
education	.1060926	.0031081	34.13	0.000	.0999969	.1121883
1.female	-.4287826	.0568614	-7.54	0.000	-.5402995	-.3172657
1.urban	.3048896	.0319092	9.55	0.000	.2423092	.3674701
female#urban						
1 1	.1307768	.0647637	2.02	0.044	.0037619	.2577916
_cons	7.981337	.0631525	126.38	0.000	7.857482	8.105191

88 . margins female#urban

Predictive margins

Number of obs = 1,919

Model VCE: Robust

Expression: Linear prediction, predict()

	Delta-method					
	Margin	std. err.	t	P> t	[95% conf. interval]	
female#urban						
0 0	9.710715	.02834	342.65	0.000	9.655135	9.766296
0 1	10.0156	.015038	666.02	0.000	9.986112	10.0451
1 0	9.281932	.0494054	187.87	0.000	9.185038	9.378826
1 1	9.717599	.0277914	349.66	0.000	9.663094	9.772103

89 .
end of do-file

90 .